MIL-T-19029E(OS)
27 August 1970
Superseding
MIL-T-19029D(OS)
31 May 1966

MILITARY SPECIFICATION

TIMER

(For Torpedo Mark 37 Mods 0 and 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of a timer, NAVORD Drawing 877432.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-Z-325	Zinc Coating (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection for
MIL-P-8585	Primer; Coating, Zinc Chromate, Low Moisture Sensitivity
MIL-A-8625	Anodized Coatings for Aluminum and Aluminum Alloys

FSC 1355

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MIL-W-16878	Wire; Electrical, Insulated, High Temperature
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base

DRAWINGS

Naval Ordnance Systems Command

NAVORD LD 620039	Control Unit Assembled (Mod 0)
NAVORD LD 620297	Control Unit Assembled (Mod 3)
NAVORD LD 495872	Shipping (Packing) Condition (For One Timer)
NAVORD Drawing 812303	High Voltage Relay
NAVORD Drawing 815753	Relay
NAVORD Drawing 877432	Timer
NAVORD Drawing 1563738	Relay, Armature

PUBLICATIONS

Naval Ordnance Systems Command

OS 6341	Miscellaneous General Ordnance Design Requirements

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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National Bureau of Standards Publication

Handbook H28 — Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20360)

3. REQUIREMENTS

3.1 Description. The timer, mounted in the control unit, NAVORD LD 620039 or LD 620297, consists of a 400 hertz (Hz) 115 volt root mean square (rms), 3-phase motor which drives, through appropriate gearing, four sets of cam-actuated switches and a continuous type potentiometer. A cover enclosing the timer can be removed to permit its adjustment.

3.2 General requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the timer shall be manufactured using the methods and procedures proposed for the production. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing timers that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the timer shall conform to the detailed requirements indicated on NAVORD Drawing 877432 and to this specification.

3.2.3.1 Standard parts and commercial items. The greatest practicable use commensurate with design requirements shall be made of standard parts. Such parts shall be selected from Military Standards and Specifications, in that order of preference.

3.2.3.2 Hookup wire. All hookup wire used in the timer shall conform to MIL-W-16878, unless otherwise specified.

3.2.3.3 Mercury and silicones. Materials capable of giving off either mercury or silicone vapor, when subjected to temperatures within the operating range of the timer, shall not be used.

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3.2.4 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for the timer shall be in accordance with the applicable drawings, specifications, and publications listed in NAVORD Drawing 877432.

3.2.4.1 Aluminum alloy parts. Aluminum and aluminum alloys, if used, shall be anodized in accordance with MIL-A-8625, Type I.

3.2.4.2 Magnesium alloy parts. Magnesium and magnesium alloys, if used, shall be treated in accordance with MIL-M-3171, Type III, followed by priming in accordance with MIL-P-8585, two coats.

3.2.5 Moisture, fungus, and corrosion. The timer shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified in the drawings and specifications listed in LD 620039 or LD 620297. When moisture, fungus, and corrosion resistance are not specified in these documents, the item shall conform to the requirements of OS 6341.

3.2.5.1 Ferrous metal parts. Electrical sheet-steel assemblies shall be processed by accepted commercial methods to provide protection against corrosion. Where other noncorrosion resistant ferrous metals must be used, the exposed surfaces shall be zinc plated in accordance with QQ-Z-325, Type II, Class 2.

3.2.6 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in Drawing 877432.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Conflicting requirements. Conflicting requirements arising between this specification or any specification, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.3 Performance requirements and product characteristics. The timer shall meet the following performance requirements and product characteristics:

3.3.1 Performance requirements

3.3.1.1 Motor input

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(a) Voltage. The input voltage shall be 3-phase, 115 volts rms plus or minus 3 percent average line-to-line voltage, with the difference between any two line voltages not exceeding 12 volts rms.

(b) Frequency. The input frequency shall be 400 Hz plus or minus 2 percent.

(c) Power. The input power to the timer motor, when the motor is operating the steady state mechanical load imposed by the switches and potentiometer specified herein, shall not exceed 20 watts.

3.3.1.2 Starting response time. The timer motor shall have sufficient starting torque to ensure that, from every possible rest position of the cam shaft, the timer will reach its steady-state operating speed within five seconds after the following applicable voltage is applied:

Three phase, 90 volts rms average line-to-line voltage maximum, 400 Hz plus or minus 2 percent. The difference between any two line-to-line voltages shall not exceed 9 volts. Steady-state operation shall be considered to exist if circuit 1 repeat-cycles in 1250 milliseconds (ms) plus or minus 27 ms.

3.3.1.3 Switch circuits

3.3.1.3.1 Repeat-cycle time. With the timer motor operating on an input frequency of 400 Hz and the applicable input voltage specified in 3.3.1.1 (a), the time required to complete one cycle of operation in each circuit listed in Table I shall be 1250 ms plus or minus 27 ms. For corrections in time due to variations in frequency, see 4.5.5.3.3.

3.3.1.3.2 Switching functions. With the timer motor operating on an input frequency of 400 Hz and the applicable input voltage specified in 3.3.1.1 (a), the timer shall perform the switching functions specified in Table I. Unless otherwise specified, all times listed in Table I are with respect to the closure of circuit 1. For corrections in time due to variations in frequency, see 4.5.5.3.3.

3.3.1.3.3 Contact ratings. Switches shall be capable of making and breaking the load current specified in Table I for the circuit in which they are used, at a voltage of 26 volts dc plus or minus 10 percent.

3.3.1.3.4 Switch circuit voltage drop. Switch circuit voltage drop shall not exceed two volts dc at the current specified in Table I when measured at the connector terminals.

3.3.1.3.5 Contact bounce. The duration of contact bounce occurring in any circuit at switch closure shall not exceed one ms. For purposes

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of time measurement, a circuit shall be considered closed when the switch contacts meet for the first time during each repeat-cycle.

3.3.1.4 Speed of potentiometer wiper arm. With the timer motor energized as specified in 3.3.1.1, the potentiometer wiper arm cycle period shall be 20 seconds plus or minus 0.7 second.

3.3.1.5 Dielectric requirements

3.3.1.5.1 DC circuits. All switches shall be capable of withstanding 500 volts rms, 60 Hz between their terminals and the timer chassis, with the motor and potentiometer terminals grounded to the chassis. Test voltage shall start at zero, be raised to 500 volts rms within 20 seconds, and then be maintained for one minute. No arcing or breakdown of insulation shall occur.

TABLE I - SWITCHING DATA

Circuit	Closes (MS)	Opens (MS)	Load
1	0	52 ±2	Coil of Relay Drawing 812303
2	625 ±6	52 ±2 after circuit 2 closes	Coil of Relay Drawing 812303
3	10 ±3 after circuit 1 opens	52 ±2 after circuit 3 closes	Coil of Relay Drawing 1563738
4	30 ±10	445 ±10	Coil of Relay Drawing 815753

3.3.1.5.2 AC circuits. The motor and potentiometer shall be capable of withstanding 1000 volts rms, 60 Hz between their terminals and the timer chassis, with all switch terminals grounded to the chassis. Test voltage shall start at zero, be raised to 1000 volts rms within 20 seconds, and then be maintained for one minute. No arcing or breakdown of insulation shall occur.

3.3.1.6 Operating time. The timer shall be capable of operating for 30 minutes plus or minus 5 minutes, followed by sufficient time to cool to ambient temperature.

3.3.1.7 Operating life. The timer shall have a useful life of at least 200 hours without lubrication or overhaul.

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3.3.2 Product characteristics

3.3.2.1 Adjustment. The timer shall be so designed that the sequence of circuit switching of each circuit can be continuously adjustable through 360 degrees. The length of switch closure time of circuits 1, 2, and 3 shall be continuously adjustable over the range from 25 to 150 milliseconds. The length of switch closure time of circuit 4 shall be continuously adjustable over the range from 315 to 515 milliseconds. The adjusting means shall be so designed that the timer will not change its setting when subjected to the conditions specified in 3.4.

3.3.2.2 Potentiometer. The potentiometer shall be capable of continuous rotation and shall have the following chsracteristics:

(a) Wattage rating. Three watts minimum, continuous at 54 degrees centigrade (C).

(b) Continuity. Wiper shall maintain contact with winding through 360 degrees of rotation.

(c) Tap points. The potentiometer shall have four tap points on its winding, as specified in 3.3.2.2 (d). Resistance between tap points shall be 5000 ohms plus or minus 5 percent.

(d) Tap positions. Zero degrees; 90 degrees plus or minus 2 degrees; 180 degrees plus or minus 2 degrees; 270 degrees plus or minus 2 degrees. The end of the winding shall be connected to the start of the winding at the zero degrees tap. The end of the winding shall be within 2 mechanical degrees of the start of the winding.

(e) Linearity. Plus or minus one percent.

(f) Resolution. A minimum of 1000 turns of wire per potentiometer.

(g) The potentiometer shall be capable of withstanding 1000 volts rms, 60 Hz between its terminals and its chassis mount.

(h) The potentiometer shall be capable of withstanding any environmental condition that may result from testing the timer as specified in 3.4.

3.3.2.3 Internal conditions. There shall be no electrical connections made between the motor circuit, any of the switch circuits, and the potentiometer circuit. No circuit within the timer shall be connected to the timer frame.

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3.4 Environmental requirements. The timer shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.4

3.4.1 Vibration. The timer shall meet the following vibration requirements:

3.4.1.1 Vibration, nonoperating. The timer shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.1.2.

3.4.1.2 Vibration, operating. The timer shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.4.1.3.

3.4.2 Temperature and humidity

3.4.2.1 High temperature and humidity, nonoperating. The timer shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.1.

3.4.2.2 High temperature, operating. The timer shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.2.2.

3.4.2.3 Low temperature, nonoperating. The timer shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.3.

3.4.2.4 Low temperature, operating. The timer shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.2.4.

3.5 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.6 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all the requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, impregnation of coils, plating, painting, machine screw

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assemblage, welding, riveting, brazing, and freedom of parts from all burrs and sharp edges.

The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the timer shall be classified as follows:

(a) Preproduction inspection (paragraph 4.4)

(b) Quality conformance inspection (paragraph 4.5)

(c) Periodic production inspection (paragraph 4.4)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five timers, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the timers by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.5, a random sample of five timers will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and draw-

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ings. The timers shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.4.1 Vibration.

4.4.1.1 Mounting. The timer, less pack or packaging, shall be mounted rigidly to a vibration device by means of its four mounting holes, with the mounting base horizontal to the vibration device, and tested as specified in 4.4.1.2 and 4.4.1.3.

4.4.1.2 Vibration, nonoperating. The timer shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.4.1.1.

4.4.1.3 Vibration, operating. The timer shall be subjected to the "Vibration, operating" test of MIL-E-82590(OS) and during and thereafter shall meet the requirements of 3.4.1.2.

4.4.2 Temperature and humidity

4.4.2.1 High temperature and humidity, nonoperating. The timer shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.4.2.1.

4.4.2.2 High temperature, operating. The timer shall be subjected to the "High temperature, operating" test of MIL-E-82590(OS) and, while at the test temperature, shall meet the requirements of 3.4.2.2.

4.4.2.3 Low temperature, nonoperating. The timer shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.4.2.3.

4.4.2.4 Low temperature, operating. The timer shall be subjected to the "Low temperature, operating" test of MIL-E-82590(OS) and, while at the test temperature, shall meet the requirements of 3.4.2.4.

4.4.3 Motor input power. Apply the applicable input voltage specified in 3.3.1.1 (a) to the timer motor. The power drawn by the timer motor when driving both the switch and the potentiometer loads shall not exceed the limit specified in 3.3.1.1 (c).

4.4.4 Starting response time. Connect the coil of Relay 812303 in series with timer circuit 1. Connect the remaining circuit 1 timer terminal and the remaining terminal on the relay coil across a 26 volt dc plus

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or minus 10 percent power supply. Record the operation of circuit 1 from the time the applicable voltage and frequency specified in 3.3.1.2 is applied to the timer motor until steady-state operation occurs in circuit 1. Repeat this test 24 times. Advance the cam shaft rest position approximately 15 degrees, for each successive test, from the rest position it had at the start of the preceding test. The timer shall perform as specified in 3.3.1.2.

4.4.5 Switch contact tests

4.4.5.1 Contact voltage drop. Connect the coil of Relay 812303 to the common terminal of circuits 1 and 2. Connect the coil of Relay 1563738 in series with circuit 3. Connect the coil of Relay 815753 in series with circuit 4. Join the remaining terminals of all relay coils together and connect them to the negative terminal of the 26 volt dc power supply. Join the remaining switch circuit terminals together and connect them to the positive terminal of the 26 volt dc power supply. Apply the applicable voltage and frequency specified in 3.3.1.1 to the timer motor. The voltage drop across each switch in the timer, while the switch is carrying load current, shall not exceed the limits specified in 3.3.1.3.4.

4.4.5.2 Contact bounce. Using the identical test setup specified in 4.4.5.1, measure the duration of contact bounce which may occur in the closure of each circuit. Contact bounce may be observed and measured by placing the input terminals of an oscilloscope, capable of driven sweep operation, across either the switch being tested or the load impedance connected in series with the switch. A standard one kHz per second oscillator may be used to calibrate the oscilloscope. The duration of the contact bounce in any timer circuit shall not exceed the limit specified in 3.3.1.3.5.

4.4.6 Potentiometer test. The potentiometer shall be checked for compliance with 3.3.2.2.

4.4.7 Operating life. For this test there may be used a new timer or a timer that has been under test and overhauled. Use the test setup specified in 4.4.5.1, and in addition, connect a source of 90 volts rms plus or minus 3 percent, 400 Hz plus or minus 2 percent to the voltage supply terminals of the potentiometer. Operate the timer for a total of 199 hours, always maintaining the input voltages and frequency within the specified limits. At the end of each 30 minutes plus or minus 5 minutes of operation, stop the timer. If the manufacturer so requests, the timer shall be allowed to cool to ambient temperature before it is again started and operated as described above. Once during each 10 hours of operation the timer operation shall be checked as specified below. During this test no maintenance shall be performed on the timer. While the timer is operating in the steady-state condition, performance shall be within limits specified in 3.3.1.1 (c), 3.3.1.3, and 3.3.1.4. When the timer is started, performance shall be within limits specified in 3.3.1.2.

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4.5 Quality conformance inspection. Each timer delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.5.1 Test conditions. Unless otherwise specified, the timer shall be subjected to quality conformance tests under the following conditions:

4.5.1.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18.33 degrees C to 35 degrees C).

4.5.1.2 Altitude: Normal ground.

4.5.1.3 Vibration: None.

4.5.1.4 Humidity: From 0 to 95 percent relative maximum, inclusive.

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required tests.

4.5.3 Test equipment. The following items of test equipment are required to perform the quality assurance tests set forth in this specification.

4.5.3.1 AC power supply: Three phase, 85 to 120 volts rms adjustable, 400 Hz plus or minus 2 percent, 0.5 ampere per line maximum required.

4.5.3.2 DC power supply: (a) 26 volts dc plus or minus 10 percent, 2.0 amperes maximum required; (b) 6 volts dc plus or minus 10 percent, 15 milliamperes maximum required.

4.5.3.3 AC voltmeter: 0-150 volts rms at 400 Hz, minimum impedance 3000 ohms.

4.5.3.4 DC voltmeter: 0-30 volts, minimum resistance 1000 ohms/volt.

4.5.3.5 Wheatstone bridge: 0-9999 ohms minimum, in steps of one ohm maximum.

4.5.3.6 Electronic counter.

4.5.3.6.1 Frequency-measurement section.

(a) Range: 390 to 410 Hz minimum.

(b) Accuracy: Plus or minus 0.1 Hz minimum.

(c) Allowable input-signal voltage: 120 volts rms. This input signal voltage may be reduced, by means of a potentiometer, to a level suitable for any otherwise acceptable commercial counter. If a potentio-

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4.5.3.6.2 Time interval measurement section.

(a) Range: 0.1 ms to 1500 ms minimum.

(b) Accuracy: Plus or minus 0.1 ms minimum.

(c) Triggering circuits. The "start of count" signal channel and the "stop counting" signal channel shall be capable of either independent or common operation. The change from independent to common operation shall be made by means of a switch.

(1) Signal range: Both the start and stop channels shall be capable of operating on signals having a positive slope and a magnitude of plus or minus 30 volts dc. The triggering level shall be continuously adjustable from plus 30 volts dc to minus 30 volts dc minimum.

(2) Input impedance: 30,000 ohms shunted by 100 micromicrofarads minimum.

4.5.3.6.3 Period measurement section.

(a) Range: 18 seconds to 22 seconds minimum.

(b) Accuracy: Plus or minus 0.01 second minimum.

(c) Allowable input signal voltage: 15 volts peak.

(d) Input impedance: 50,000 ohms shunted by 100 micromicrofarads minimum.

4.5.3.7 Four dummy loads, constructed as follows:

4.5.3.7.1 One resistive leg shall be connected in parallel with a capacitive-resistive leg. The junction between the two resistors shall be called terminal A.

4.5.3.7.2 The remaining connection between the parallel legs shall be called terminal B.

4.5.3.7.3 The connection between the resistor and capacitor in the capacitive-resistive leg shall be called terminal C.

(a) Resistive leg: 330 ohms plus or minus 10 percent, 2 watt resistor.

(b) Capacitive-resistive leg: 2 microfarads plus 20 percent minus 10 percent, paper dielectric, 50 volts dc minimum, capacitor, connected in series with a 3300 ohm plus or minus 10 percent, 1/2 watt resistor.

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4.5.3.8 High potential test set. This set shall be capable of furnishing 500 volts rms and 1000 volts rms at 60 Hz. The set will be used for insulation testing, and shall include an alarm bell or light or combination of both to indicate when arcing or breakdown occurs.

4.5.3.9 Switch. One three-pole, single-throw switch is required. The switch shall have sufficient insulation resistance and creepage distance between adjacent poles to withstand 1000 volts rms at 60 Hz for one minute without any arcing or breakdown of insulation. Each switch pole shall be capable of carrying a one-half ampere minimum reactive load current at 120 volts rms, 400 Hz continuously without any evidence of overheating. Maximum contact drop at one-half ampere rms shall not exceed 0.1 volt rms.

4.5.3.10 Recorder. Sonborn Model 151 single channel dc amplifier recorder or equivalent.

4.5.4 Test setup

4.5.4.1 Lock the circuit-setting adjustments in place.

4.5.4.2 Using a separate dummy load for each connection, connect the B terminal of a dummy load to each of the following: P152 terminal 9, P152 terminal 11, a terminal of circuit 3, and a terminal of circuit 4.

4.5.4.3 Connect the A terminals of all dummy loads to the negative terminal of the 26 volt dc power supply.

4.5.4.4 Join terminal 10 and the remaining terminals of circuits 3 and 4 together and connect that to the positive terminal of the 26 volt dc power supply.

4.5.4.5 Connect the timer motor input terminal to the load side of the three-pole, single-throw switch. The switch shall be in the open position. Connect the input side of the switch to the 85-120 volt ac power supply.

4.5.5 Test procedure. (See NAVORD Drawing 877432 for timer terminal designations.)

4.5.5.1 Voltage measurements

4.5.5.1.1 Motor input voltage. Using the ac voltmeter, set the ac power supply voltage as specified in 3.3.1.2. Close the three-pole single-throw switch. The timer shall start functioning without additional external aid. Open switch and repeat the above procedure a total of four times, all at different initial cam positions. Reset the motor input voltage to the value specified in 3.3.1.1.

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4.5.5.1.2 Switch circuit voltage. Using the dc voltmeter, measure and record the voltage supplied to the timer switch circuits. It shall be as specified in 3.3.1.3.3.

4.5.5.2 Frequency. Using the frequency measuring section of the electronic counter, measure and record the timer motor input frequency. It shall be as specified in 3.3.1.1.

4.5.5.3 Switch circuit time measurements

4.5.5.3.1 Connect the ground terminal of the time interval section of the electronic counter to the negative terminal of the dc power supply. Connect the "start of count" and "stop counting" leads of the electronic counter to terminal C only of the applicable dummy load for each test specified in Table II. Where positive triggering voltage polarities are specified in Table II, the electronic counter trigger slope setting shall be positive. Where negative triggering voltage polarities are specified in Table II, the electronic counter trigger slope settings shall be negative. Triggering levels on the electronic counter shall be three plus or minus one volt dc less than the the voltage measured in 4.5.5.1.2. Measure and record the maximum and minimum time intervals for at least four cycles, as read by the electronic counter, for each test specified in Table II.

TABLE II - ELECTRONIC COUNTER SETTINGS

	"Start of count" Channel		"Stop counting" Channel		
Test	Connect to	Trigger Polarity	Connect to	Trigger Polarity	Channel Operation
1	circuit 1	positive	- - - - -	negative	common
2	circuit 1	positive	circuit 2	positive	independent
3	circuit 2	positive	- - - - -	negative	common
4	circuit 1	negative	circuit 3	positive	independent
5	circuit 3	positive	- - - - -	negative	common
6	circuit 1	positive	circuit 4	positive	independent
7	circuit 1	positive	circuit 4	negative	independent
8	circuit 1	positive	- - - - -	positive	see 4.5.5.3.2

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4.5.5.3.2 Set channel operation to "independent." Reset the electronic counter so that zero time is read. Within 0.5 second maximum after the electronic counter begins to count due to the application of the "start of count" triggering voltage, switch channel operation to "common."

4.5.5.3.3 Test data correction. The data taken as specified in Table II shall be corrected for timer motor input frequency as follows:

$$\text{Corrected reading} = \left(1 + \frac{\text{Frequency of } 4.5.5.2 - 400 \text{ Hz}}{400 \text{ Hz}}\right) \times \text{actual reading}$$

4.5.5.3.4 Test results. The frequency-corrected results of 4.5.5.3.3 shall be as follows. The data listed below are taken from Table I and 3.3.1.3.1, and repeated here for convenience.

Test 1.	52 ±2 ms	Test 5.	52 ±2 ms
Test 2.	625 ±6 ms	Test 6.	30 ±10 ms
Test 3.	52 ±2 ms	Test 7.	445 ±10 ms
Test 4.	10 ±3 ms	Test 8.	1250 ±27 ms

4.5.5.4 Potentiometer operation

4.5.5.4.1 Connect terminal 4 of P152 to the positive terminal of the 6 volt dc power supply. Connect terminal 5 of P152 to the negative terminal of the 6 volt dc power supply. Connect terminal 6 of P152 to the ground terminal of the recorder. Connect terminal 7 of P152 to the input terminal of the recorder. Set the recorder chart speed to 25 mm/sec, and set the gain to 0.2 volt/mm.

4.5.5.4.2 Record at least two complete cycles of potentiometer operation. It shall be as specified in 3.3.1.4.

4.5.5.4.3 Observe the rise and fall of the potentiometer output voltage. The voltage variation shall not be erratic.

4.5.5.5 Potentiometer resistance. Using the Wheatstone bridge, measure the potentiometer resistance between terminals 4 and 6 and 5 and 6 on P152. Each reading shall be 2500 ohms plus or minus 5 percent.

4.5.5.6 Dielectric test. Apply test voltages to the timer terminals, as specified in 3.3.1.5. No arcing or breakdown of insulation shall occur.

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4.5.5.7 Examine timer design to determine that adjustment requirements and internal conditions are in accordance with 3.3.2.1 and 3.3.2.3.

4.5.5.8 Visual and dimensional examination. Each timer shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.5.5.9 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The timer shall be packaged in accordance with Method IID of MIL-P-116 and LD 495872.

5.1.2 Level B. Not applicable

5.1.3 Level C

5.1.3.1 Unit packaging. The timer shall be packaged in accordance with Method III of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C as specified (see 6.2).

5.2.1 Number of packages per pack. The number of unit packages per pack, for Level C, shall be as specified in the contract or order.

5.2.2 Level A

5.2.2.1 Unit pack. The timer shall be packed in accordance with LD 495872 and may be shipped individually in small lots.

5.2.2.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Load Type III (see 6.2).

5.2.3 Level B. Not applicable

5.2.4 Level C. The timer shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

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5.3 Marking

5.3.1 Normal markings, Level A. Unit packs shall be marked in accordance with the requirements of MIL-STD-129 and LD 495872.

5.3.2 Normal markings, Level C. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.3 Serial numbers. Unit packages and shipping containers shall be marked with the serial numbers of the units contained within.

6. NOTES

6.1 Intended use. The timer is intended for use in Torpedo Mk 37 Mods 0 and 3.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) Data required other than that specified in 3.2.1.

(g) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.3 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

Level C. This level shall be used only when it is definitely known that the packaged items are to be shipped to domestic installations for immediate use at the first receiving activity.

Preparing Activity
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